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College of Informatics and Computing Sciences

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

COURSE INFORMATION SYLLABUS (CIS)

Vision	A premier national university that develops leaders in the global knowledge economy.			
Mission	A university committed to producing leaders by providing a 21st century learning environment through innovations in education, multidisciplinary research, and community and industry partnerships in order to nurture the spirit of nationhood, propel the national economy and engage the world for sustainable development.			
Course Title	Introduction to Computing		**Course Code**	CC 100
Course Category	Professional Course		**Pre-requisite(s)**	None
Semester/Year	First Semester/First Semester		**Credit Hours**	5 hours
Course Instructor	Zymon Andrew M. Maquinto	56467	**Reference CMO**	CMO 5. Series 2015
	Master of Science in Data Science, Doctor of Jurisprudence		**Date Prepared**	August 5, 2026
	zymonandrewm.maquinto@g.batstate-u.edu.ph		**Revision No.:**	0
Period of Study	Academic Year 2026-2027		**Revision Date:**	
Course Rationale and Description	Introduction to Computing presents practical and outcome-based approaches and techniques toward understanding the core principles, architectural frameworks, and societal impacts of modern computing systems. The course highlights the historical evolution of hardware, software abstractions, network infrastructure, algorithmic problem-solving tools, and data representation methodologies. The specialized application of these techniques is geared on extended learning in design and development of digital solutions with focus on introductory topics in computing and emerging technologies.			
Contact Hours	2 Hours Lecture 3 Hours Laboratory			
Criteria for Assessment	20% Written Assessment 20% Laboratory Assessment 15% Preliminary Assessment 15% Midterm Examination 15% Final Examination 15% Final Project			
Teaching, Learning, and Assessment Strategies	The course shall adopt blended learning modality which utilizes both lecture and laboratory sessions. Learning modules, study guides and reviewers will be provided to students. The announcement of class activities, academic requirements and coursework shall be coursed through Facebook Messenger group chat and the LMS. The following teaching, learning and assessment strategies that will be used are as follows: **Interactive Discussions.** The class discussions will allow participation and inputs of students through discussions, demonstrations and mini-bootcamp sessions. **Written Assessments.** The students will be engaged in formative assessments, unit quizzes, module worksheets, and seatwork. These classroom activities involve pen-and-paper assessments during Lecture contact hours. **Laboratory Assessments.** The students will have laboratory activities through hands-on exercises. These laboratory tasks involve programming and configurations during Laboratory contact hours. **Major Examinations.** There will be 3 major assessments (Preliminary Exam, Midterm Exam and Final Exam) given after every 30 contact hours, one per ILO. **Final Project.** The students will be assessed on a milestone project activity as a team/group.			

Intended Learning Outcomes (ILO)	**ILO**	**Upon completion of this course, the students should be able to:**
	ILO1	Understand the evolution of computing and the areas of study in Information Technology.
	ILO2	Analyze the function and purpose of computing components and ethical considerations.
	ILO3	Acquire preparatory skills in systems design through design tools, principles and patterns.

Assessment Method and Distribution Map

Assessment Tasks (AT) Distribution				**Intended Learning Outcomes**			**Domains**		
Code	**Assessment Tasks**	**I/R/D**	**(%)**	**1**	**2**	**3**	**C**	**P**	**A**
WA	Written Assessment	R	20	25	25	25	75		
LA	Laboratory Assessment	R/D	20	30	30	30		90	
PA	Preliminary Assessment	R	15	80			80		
MA	Midterm Assessment	R	15		80		80		
FA	Final Assessment	R	15			80	80		
FP	Final Project	R/D	15			50		30	20
	Total		**100**				**315**	**120**	**20**
Note:	All internal assessments with feedback will be made available within 2 week after each assessment submission except Final Project.								

Books and Other References	1	Congzon, V. C. (2025). Introduction to computing. Mindshapers Co., Inc.
	2	Falcon, I. G. (2025). Introduction to computing & ICT 1. Mindshapers Co., Inc.
	3	Farrell, J. (2026). Programming Logic and Design Introductory Philippine Edition, Course Technology for Cengage Learning Publishing
	4	Abante, M. V., Bacay, J. A., Julianes, M. T., & Tiamzon, R. A. (2026). Legal, social, and professional ethics in computing education. Mindshapers Co., Inc.
	5	Cashman, S. et al. (2007). HTML Comprehensive Concepts and Techniques, Comprehensive Fourth Edition, Thomson Course Technology Series.
Institutional Graduate Attributes (IGA)	**IGA**	**Institutional Graduate Attributes (IGA) Statements**
	IGA1	**Knowledge Competence**
		Demonstrate a mastery of the fundamental knowledge and skills required for functioning effectively as a professional in the discipline, and an ability to integrate and apply them effectively to practice in the workplace.
	IGA2	**Creativity and Innovation**
		Experiment with new approaches, challenge existing knowledge boundaries and design novel solutions to solve problems.
	IGA3	**Critical and Systems**
		Identify, define, and deal with complex problems pertinent to the future professional practice or daily life through logical, analytical and critical thinking.
	IGA4	**Communication**
		Communicate effectively (both orally and in writing) with a wide range of audiences, across a range of professional and personal contexts, in English and Pilipino.
	IGA5	**Lifelong Learning**
		Identify own learning needs for professional or personal development; demonstrate an eagerness to take up opportunities for learning new things as well as the ability to learn effectively on their own.
	IGA6	**Leadership, teamwork, and Interpersonal Skills**
		Function effectively both as a leader and as a member of a team; motivate and lead a team to work towards goal; work collaboratively with other team members; as well as connect and interact socially and effectively with diverse culture.
	IGA7	**Global Outlook**
		Demonstrate an awareness and understanding of global issues and willingness to work, interact effectively and show sensitivity to cultural diversity.
	IGA8	**Social and National Responsibility**
		Demonstrate an awareness of their social and national responsibility; engage in activities that contribute to the betterment of the society; and behave ethically and responsibly in social, professional and work environments.
Student Outcomes (SO)	**SO**	**Student Outcomes (SO) Statements**
	SO1	**Discipline Knowledge**
		Ability to analyze complex computing problems and to apply principles of computing and other relevant disciplines to identify solutions.
	SO2	**Design/Development of Solutions**
		Ability to design, implement, and evaluate computing-based solutions to meet a given set of computing requirements in the context of the program's discipline.
	SO3	**Communication**
		Ability to communicate effectively in a variety of professional contexts.
	SO4	**Ethics and Professionalism**
		Ability to recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.
		Individual and Teamwork
	SO5	Ability to function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.
		Systemic Thinking
	SO6	Use systemic approaches to select, develop, apply, integrate, and administer secure computing technologies to accomplish user goals.
CDIO Framework Skills	**CDIO**	**CDIO Skills**
	CDIO1	**Disciplinary Knowledge & Reasoning**
		Knowledge of underlying mathematics and sciences, core engineering fundamental knowledge, advanced engineering fundamental knowledge, methods and tools
	CDIO2	**Personal and Professional Skills & Attributes**
		Analytical reasoning and problem solving; experimentation , investigation and knowledge discovery; system thinking; attitudes, thoughts and learning; ethics, equity and other responsibilities
	CDIO3	**Interpersonal Skills: Teamwork & Communication**
		Teamwork, communications, communication in a foreign language
	CDIO4	**Conceiving, Designing, Implementing & Operating Systems**
		External, societal and environmental context, enterprise and business context, conceiving, systems engineering and management, designing, implementing, operating

Sustainable Development Goals Skills	SDG	SDG Skills
	SDG1	**Envisioning**
		Establish a link between long-term goals and and immediate actions, and motivate people to take action by harnessing their deep aspirations.
	SDG2	**Critical Thinking and Reflection**
		Examine economic, environmental, social and cultural structures in the context of sustainable development, and challenges people to examine and question the underlying assumptions that influence their world views by having them reflect on unsustainable practices.
	SDG3	**Systemic Thinking**
		Recognise that the whole is more than the sum of its parts, and it is a better way to understand and manage complex situations.
	SDG4	**Building Partnerships**
		Promote dialogue and negotiation, learning to work together, so as to strengthen ownership of and commitment to sustainable action through education and learning.
	SDG5	**Participation in Decision Making**
		Empower oneself and others through involvement in joint analysis, planning and control of local decisions.

COURSE POLICIES

A. GRADING SYSTEM

The grading system adopted by this course is as follows:

Excellent	1.00	98 - 100
Superior	1.25	94 - 97
Very Good	1.5	90 - 93
Good	1.75	88 - 89
Meritorious	2.00	85 - 87
Very Satisfactory	2.25	83 - 84
Satisfactory	2.50	80 - 82
Fairly Satisfactory	2.75	78 - 79
Passing	3.00	75 - 77
Failure	5.00	Below 70
Incomplete	INC	

*Students who got a computed grade of 70-74 will be given an appropriate remedial activity in which the final grade should be either passing (3.0) or failure (5.0). No rounding off of grades shall be allowed.

B. CLASS POLICY

Prompt and regular attendance of students is required. Total unexcused absences shall not exceed ten (10) percent of the maximum number of hours required per course per semester (or per summer term). A Midterm Class semester has 18 weeks compressed to 6 weeks.

MISSED EXAMINATIONS

Students who failed to take the exam during the schedule date can be given a special exam provided he/she has valid reason. If it is health reason, he/she should provide the faculty with the medical certificate signed by the attending Physician. Other reasons shall be assessed first by the faculty to determine its validity.

ACADEMIC DISHONESTY

Academic dishonesty includes acts such as cheating during examinations or plagiarism in connection with any academic work. Such acts are considered major offenses and will be dealt with according to the University's Student Norms of Conduct.

DROPPING

Dropping must be made official by accomplishing a dropping form and submitting it at the Registrar's Office before the midterm examination. Students who officially drop out of class shall be marked "Dropped" whether he took the preliminary examination or not and irrespective of their preliminary grades.

A student who unofficially drops out of class shall be given a mark of "5.0" by the instructor.

C. OTHER COURSE POLICIES AND REQUIREMENTS

Consultation may be done from Mondays through Thursday from 8AM to 5PM. Questions may be coursed through Google Chat or Gsuite email only.

Teaching, Learning, and Assessment (TLA) Activities

Ch.	Topics / Reading List	Wks	Topic Outcomes	ILO	SO	Delivery Method
	Orientation & Introduction	1	Presentation of Syllabus, Class Rules			Discussion
1	**Evolution of Computing** *Brief History of Electronic Computing* *Fourth Industrial Revolution* *Internet and Network Technologies* *Mobile Computing Revolution* *Artificial Intelligence* *Modern Frontiers in Computing*	1-2	Analyze the evolution of electronic computing, and evaluate how the convergence of network technologies, mobile ubiquity, and artificial intelligence drives automated ecosystem design	1	2	Lecture, Demonstration and Interactive Discussions
	Written Assessment #1	2		1	2	Quiz

2	**Areas of Computing** *Information Technology vs Computer Science* *Software Engineering and Application Development* *Big Data and Blockchain* *Cloud Computing* *Machine Learning* *Internet of Things (IoT) and Smart Cities* *Cybersecurity* *Augmented Reality and Creative Computing* *Service Management, BPO and VA*	3-4	Understand the different areas of computing representing different emerging technologies and methodologies	1	2	Lecture, Demonstration and Interactive Discussions
	Laboratory Assessment #1	4		1	2,5	Skills Check
3	**Preliminary Assessment**	5		1	2	Written Test
4	**Components of Computing** *Hardware and Devices* *Platforms and Operating Systems (OS)* *Applications and Utilities* *Network and Communication* *Data and Information* *Number Systems*	6-7	Analyze the components of computing involving hardware, software and network	2	2	Lecture and Interactive Discussions
	Laboratory Assessment #2	7		2	2,5	Skills Check
5	**Ethics of Computing** *Foundations of Computer Ethics* *Professional Codes of Conduct* *Data Privacy* *Cybersensorship* *Algorithmic Bias and Discrimination* *Intellectual Property* *Workplace and Economic Disruptions* *Environmental and Global Impact*	8	Appreciate the importance of ethical judgment and decision-making in computing applications, tasks and profession in general	2	2	Lecture, Demonstration and Interactive Discussions
	Written Assessment #2	8		2	2	Quiz
6	**Midterm Assessment**	9		2	2	Written Test
7	**Design Tools** *System Modeling Fundamentals* *Conditional and Iterative Structures* *Flowchart* *ERD Chen and Crow's Foot Notations* *Pseudocode Algorithm* *Unified Modeling Language (UML)*	10-12	Apply design tools such as flowchart, ERD and pseudocode in real-world scenarios, use cases and computing problems	3	2	Lecture, Demonstration and Interactive Discussions
	Written Assessment #3	12		3	2	Practice Sets
8	**Design Principles** *SOLID Object-Oriented Design* *GRASP Patterns* *Programming Paradigms* *Core Coding Philosophies* *Architectural Design Principles* *UI/UX* *Security by Design (SbD)*	13-14	Understand the different design principles as applied to systems analysis, design and development	3	2	Lecture, Demonstration and Interactive Discussions
	Laboratory Assessment #3	14				Skills Check
	Design Patterns *Gang of Four (GoF)* *Pattern Template* *Creational Design Patterns* *Structural Design Patterns* *Behavioral Design Patterns*	15-16		3		Lecture, Demonstration and Interactive Discussions
	Final Project	17			2,5	Task Performance
	Final Assessment	17		3	2	Written Test

Assessment Schedule		**Week No.**														
	Distribution	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**
Assessment Method	Written Assessment	X						X				X					
	Laboratory Assessment			X			X							X			
	Preliminary Assessment				X												
	Midterm Examination								X								
	Final Examination																X
	Final Project																X

	ILOs	STUDENT OUTCOMES (SO): Mapping of Assessment Tasks (AT)								
		2	5					C	P	A
ILO-SO and ILO-CPA Mapping	ILO1	WA, LA, PA	LA					WA, PA	LA	
	ILO2	WA, LA, MA	LA					WA, MA	LA	
	ILO3	WA, LA, FA, FP	LA, FP					WA, FA	LA, FP	FP
	ILO4									
	ILO5									
	ILO6									
	ILO7									

	ILOs	INSTITUTIONAL GRADUATE ATTRIBUTES (IGA): Mapping of Assessment Tasks (AT)								
		2	4	6						
ILO-IGA Mapping	ILO1	WA, PA	LA	LA						
	ILO2	WA, MA	LA	LA						
	ILO3	WA, FA	LA, FP	LA, FP						
	ILO4									
	ILO5									
	ILO6									
	ILO7									

	ILOs	CDIO SKILLS				SDG Skills				
		4				3	5			
ILO-CDIO and ILO-SDG Mapping	ILO1	WA, LA, PA				WA, PA				
	ILO2	WA, LA, MA				WA, MA	LA			
	ILO3	WA, LA, FA, FP				WA, FA	LA, FP			
	ILO4									
	ILO5									
	ILO6									
	ILO7									

Prepared by:	Reviewed by:	Approved by:
ZYMON ANDREW M. MAQUINTO Faculty	**DIONECES O. ALIMOREN** Program Chairperson, CICS	**RYNDEL V. AMORADO, DIT** Dean, CICS
Date:	Date:	Date:

Remarks:

1 The syllabus is to be distributed to the students in the first week of the semester.
2 Any changes to the syllabus shall be communicated (in writing) to the Program Chair and the approved revised version must be communicated to the students.
3 The course instructor may set a more stringent similarity percentage (minimum 20%) for their respective courses pertaining to student's submissions. However, it must be communicated in writing to the respective Program Chair and the approved revised version must be communicated to the students.